



Cumulative effects of chronic deer browsing and clear-cutting on regeneration processes in second-growth white spruce stands



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ABSTRACT

Interactions between multiple disturbances can alter resilience mechanisms, thereby triggering alternative successional pathways. Regeneration processes are important mechanisms of forest resilience because they drive successional pathways. On Anticosti Island, chronic browsing by introduced white-tailed deer (*Odocoileus virginianus*) shifted composition of understory regeneration of overmature balsam fir (*Abies balsamea*) forests toward dominance by white spruce (*Picea glauca*). Historic clear-cutting of these altered forests generated mature second-growth white spruce stands. However, the cumulative effect of chronic deer browsing and recent clear-cutting on regeneration processes of mature second-growth white spruce stands has not yet been evaluated. Our objective is to evaluate if regeneration processes would enable white spruce stands to recover from the cumulative effects of these two disturbances. We studied regeneration in relation to seed availability, substrate suitability for seedling establishment, and substrate availability in mature second-growth white spruce stands and recent clear-cuts of mature second-growth white spruce stands. Our results indicate regeneration failure in both ecosystems, which can be explained by a lack of suitable rotten logs for sufficient establishment of white spruce seedlings. Hence, the cumulative effects of chronic deer browsing and clear-cutting of mature second-growth white spruce stands have altered regeneration processes and triggered an alternative successional pathway toward parklands, i.e., partial deforestation. We propose shelterwood cuttings that create nurse logs should be investigated to maintain white spruce stands without planting.

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1. Introduction

Interactions between multiple disturbances can alter resilience mechanisms of forests, thereby triggering alternative successional pathways (Paine et al., 1998; Tremblay et al., 2007; Buma and Wessman, 2011). Resilience is the capacity of a system to absorb disturbance and reorganize so that the same structure and functions are essentially recovered, e.g., forest recovering to forest following perturbation (Holling, 1973; Gunderson, 2000). Regeneration processes are important mechanisms of forest resilience because they drive successional pathways (Buma and Wessman, 2011; Hidding et al., 2013). For example, cervids such

as white-tailed deer (*Odocoileus virginianus*) can shift the composition of understory regeneration by selectively browsing seedlings (Tremblay et al., 2007; Nuttle et al., 2014), while forest management practices can change the composition and density of regeneration through diverse silvicultural scenarios (Boucher et al., 2009; Lundmark et al., 2013; Rist and Moen, 2013). Interaction between these two disturbances could potentially alter regeneration processes, triggering alternative successional pathways that move the system toward assemblages of species that have not co-occurred historically. The occurrence of such novel ecosystems can represent a threat to biodiversity and ecosystem services that are provided by preindustrial forests (Hobbs et al., 2006; Bridgewater et al., 2011). In this context, a recent cause for concern is increasing cervid densities in the forests of North America (Côté et al., 2004; McLaren et al., 2004; Chollet and Martin, 2012), which increases the probability for potential interactions between cervid browsing and forest management.

Following the introduction of 200 white-tailed deer into the forests of Anticosti Island (Quebec) at the end of the 19th century,

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their population densities have now reached more than 20 individuals km^{-2} in the absence of predators (Rochette and Gingras, 2007). Since the 1920s, chronic deer browsing has shifted the understory composition of balsam fir forests on the island from regeneration that is dominated by balsam fir (*Abies balsamea* (L.) Mill.) towards one that is dominated by white spruce (*Picea glauca* (Moench) Voss), together with a recalcitrant ground layer that includes blue joint grass (*Calamagrostis canadensis* (Michaux) P.Beauv.), bracken fern (*Pteridium aquilinum* (L.) Kuhn), and thistle (*Cirsium* spp.), which are species that are not browsed by deer (Potvin et al., 2003; Tremblay et al., 2007; Hidding et al., 2013). Recalcitrant understory layers are dense and persistent monodominant strata that can impede regeneration, by altering the rate and shifting the direction of forest succession (Royo and Carson, 2006). Such a layer can persist even after reductions in deer densities (Nuttall et al., 2014). Historically, the unaltered forests of Anticosti Island were driven by gap disturbance and gap-filling processes that assured the maintenance of overmature balsam fir stands, mixed with white spruce and paper birch (*Betula papyrifera* Marshall; Barrette et al., 2010). Balsam fir regeneration created dense and persistent understory seedling banks, which were highly tolerant of shade and that were prompt to react and fill canopy gaps (Côté and Bélanger, 1991; McCarthy, 2001; McCarthy and Weetman, 2007a). From the 1920s onwards, white spruce has represented the sole remaining tree species that is available to recreate a forest after disturbances. In such a context, white spruce regeneration formed white spruce stands following historic clear-cutting (1910–1915) of altered balsam fir stands (Fig. 1; Beaupré et al., 2004). These now mature second-growth white spruce stands represent a threat to biodiversity and the services that were provided by the preindustrial forest (Barrette et al., 2010), since they are novel ecosystems which are not commonly found in eastern North America (Stiell, 1976; Bell et al., 1990; Lieffers et al., 2008; Saucier et al., 2009). Many mature second-growth white spruce stands have been recently clear-cut (1999) under a system

of timber production that uses even-aged management. However, the cumulative effects of chronic deer browsing and clear-cutting on regeneration processes within white spruce stands has yet to be evaluated. If regeneration processes cannot enable the recovery of white spruce stands from these two disturbances, then alternative successional pathways could develop and lead towards deforestation, since white spruce is the only remaining tree species. The occurrence of a non-forest state would represent an even greater threat to biodiversity and ecosystem services. To avoid deforestation and meet sustainable forest management objectives, forest managers would then have to resort to plantation silviculture.

Consequently we investigated if regeneration processes would enable white spruce stands to recover from the cumulative effects of chronic deer browsing and clear-cutting. To meet this objective, we studied regeneration in relation to seed availability, substrate suitability for seedling establishment, and substrate availability in mature second-growth white spruce stands and in recent clear-cuts of mature second-growth white spruce stands. We also studied regeneration processes in the understory and within canopy gaps of overmature balsam fir stands to provide a baseline of regeneration characteristics which have enabled fir stands to recover partially from historic clear-cuts and reproduce white spruce stands. We predicted that regeneration in mature second-growth white spruce stands would not be sufficient for them to recover from clear-cutting because of a lack of suitable substrates for establishment. Effectively, white spruce requires specific substrates for optimum seedling establishment (e.g., large woody debris, exposed mineral soil; Simard et al., 1998) and rarely forms such pure even-aged stands under natural conditions (Stiell, 1976; Bell et al., 1990). We also predicted that regeneration would not be sufficient in recent clear-cuts of mature second-growth white spruce stands for them to return to a forest state because of encroachment by recalcitrant understory layer species, thereby exacerbating the lack of suitable substrates for establishment.

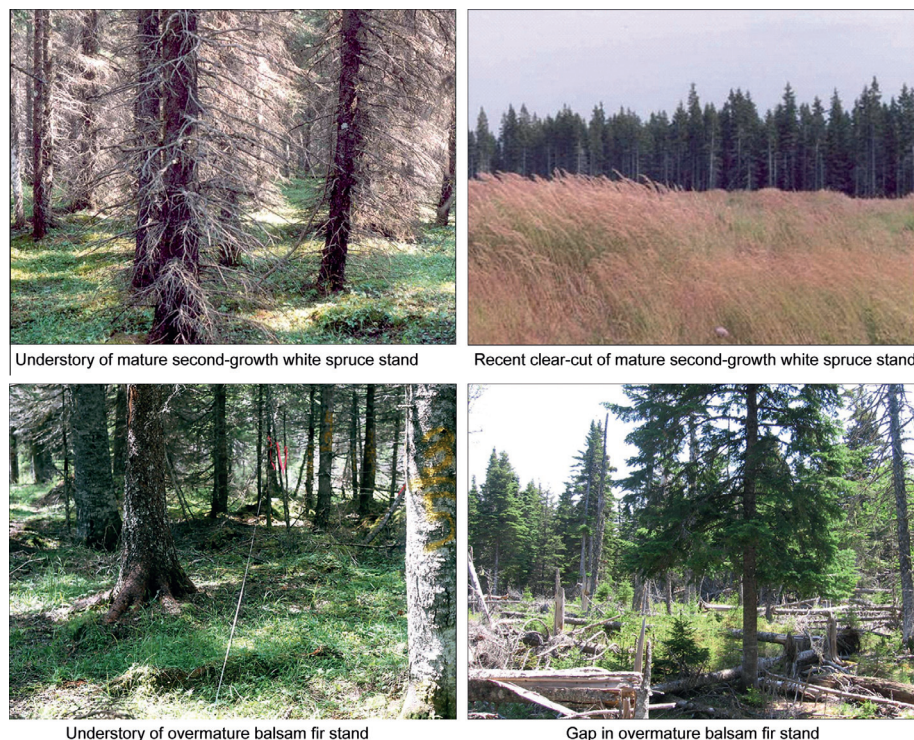


Fig. 1. Photographs showing the four ecosystem types of the study.

2. Materials and methods

2.1. Study area

Anticosti Island (7943 km²) is located in the Gulf of St. Lawrence (49°28'N 63°00'W; Fig. 2). The island is part of the eastern balsam fir-paper birch bioclimatic subdomain of Quebec (Saucier et al., 2009). Forest management on Anticosti Island is subjected to ecosystem management principles as is all crown land of Quebec, according to provincial legislation (RLRQ, c. A-18.1, article 1). Mean annual temperature is 2 °C and mean annual precipitation is 907 mm (Environment Canada, 1982). The preindustrial forest was dominated (75%) by softwood stands ≥ 100-years-old (Barrette et al., 2010). Natural disturbances include blowdowns, outbreaks of spruce budworm (*Choristoneura fumiferana* (Clem.)) and hemlock looper (*Lambdina fiscellaria* (Guen.)) defoliation

(Jobin, 1980; Martel, 1999; Chouinard and Filion, 2001) and, to a much lesser extent, fire (Lavoie et al., 2009). Sampling of mature second growth white spruce stands (~90-years-old; Fig. 1) and recent clear-cuts (6-years-old) of mature second-growth white spruce stands were restricted to the north-western portion of the island, which contains most of these two types of ecosystems (Fig. 2). Sampling of overmature balsam fir stands (>90-years-old) was restricted to the western half of the island where the concentration of such stands was highest. The eastern part of the island has a higher proportion of unbrowsed black spruce (*Picea mariana* (Miller) BSP) forests and wetlands. In overmature balsam fir forests, chronic deer browsing has not altered development of canopy trees because their crowns were beyond the reach of deer when browsing began to inhibit tree recruitment (Chouinard and Filion, 2001; Potvin et al., 2003; Barrette et al., 2010).

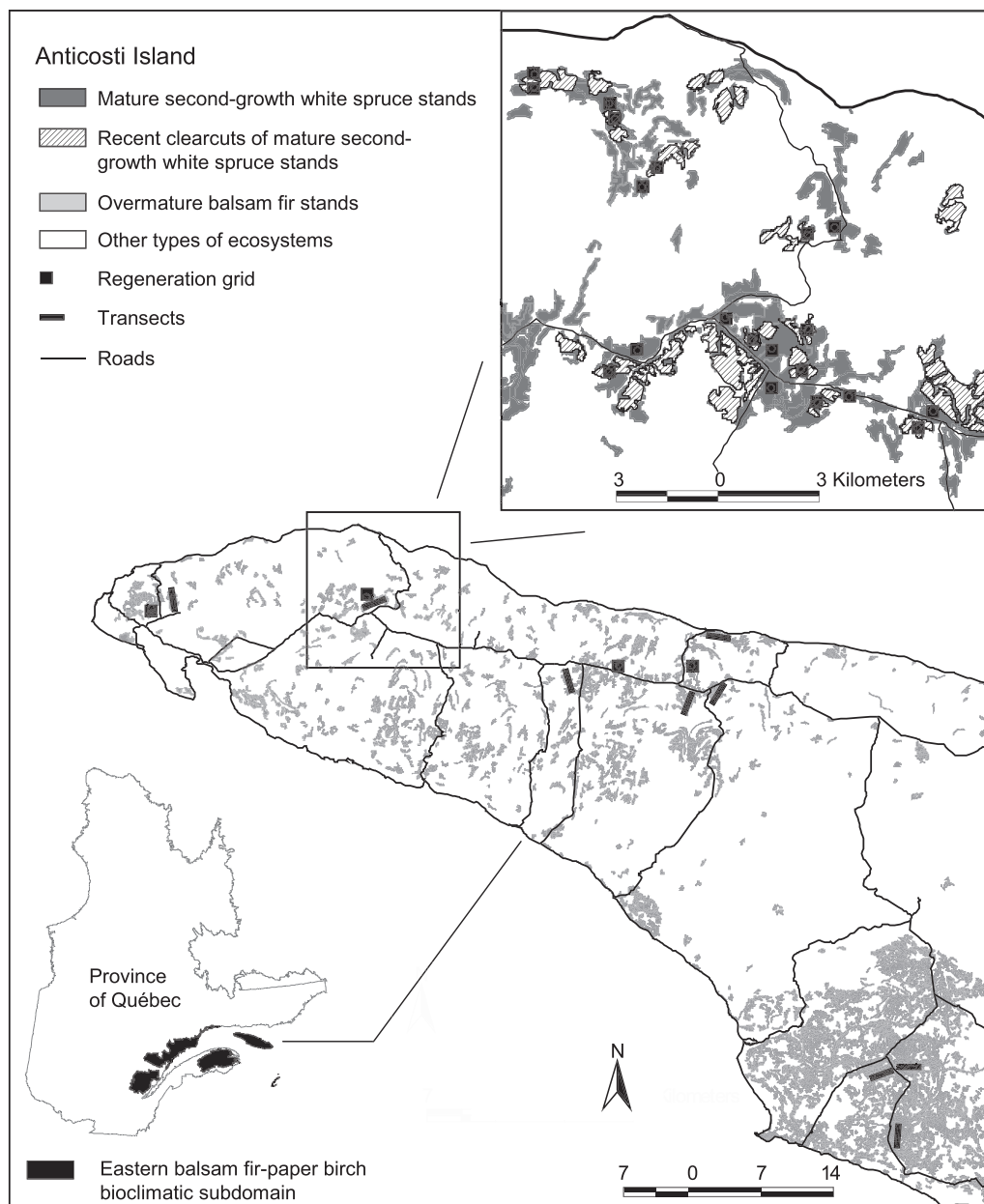


Fig. 2. Location of regeneration grids and transects in the four ecosystem types that were studied on Anticosti Island, Québec, Canada.

2.2. Field sampling

Sample site selection was based on forest inventory maps that were obtained from the Ministère des Ressources naturelles du Québec (MRNQ). To satisfy selection criteria, stands had to be situated on regional till with a slope <5%, within 5 km of a forest road, and on mesic sites in the successional series of the balsam fir–paper birch association (Saucier et al., 2009). Ten mature second-growth white spruce stands (>75% of total basal area consisting of white spruce), ten recent clear-cuts of mature second-growth white spruce stands, and thirteen overmature balsam fir stands were randomly selected from all available ecosystems of each type (Fig. 2). Four fir stands were used to study regeneration in the understory, while nine were used to study regeneration in canopy gaps ($n = 53$). We selected up to six gaps (1 out of 3) that were encountered along a transect, which had been placed along the longest axis of each of the nine fir stands.

Densities of seedlings were measured in circular subplots (4 m^2), according to 5 height classes (cotyledon, 1–10 cm, 11–30 cm, 31–60 cm, 61–130 cm). We also determined the substrate type upon which every seedling had established and visually estimated the area that was covered by these different substrates in each subplot. Forest floor was characterized according to 15 substrate types: (1) exposed mineral soil (mineral soil); (2) needles of coniferous trees (Needle); (3) leaves of deciduous broad-leaved trees (Leaf); (4) leaves of deciduous broad-leaved trees, mixed with coniferous tree needles (Needle + Leaf); (5) moss; (6) dry moss; (7) moss mixed with coniferous tree needles (Moss + Needle); (8) exposed rock (Rock); (9) logs of recently dead trees with bark still attached (Fresh Log); (10) undecomposed logs without bark (Dry Log); (11) decomposed logs (Rotten Log); (12) decomposed logs covered with moss (Rotten Log + Moss); (13) decomposed logs covered with moss and needles (Rotten Log + Moss + Needles); (14) decomposed logs covered with dry moss (Rotten Log + Dry Moss); and (15) downed dead individuals of recalcitrant understory layer species (Recalcitrant Layer Species). Heights of logs and live individuals of recalcitrant understory layer species were measured. In the middle of each selected stand and recent clear-cut, we placed 100 circular subplots, ten meters apart, within a $100 \times 100 \text{ m}$ grid. In selected gaps, we placed up to six circular subplots that were spaced 10 m apart along the transect. To describe stand characteristics and potential seed trees, we measured diameter at breast height (DBH) of live and dead saplings (DBH, 1.1–9.9 cm) and trees (DBH, $\geq 10 \text{ cm}$), which were located within a circular plot (400 m^2) that was placed in the middle of four mature second-growth white spruce stands and four overmature balsam fir stands. Seed availability was measured with 64 seed traps (0.83 m^2) that were placed in the same eight stands, and which were positioned in pairs at each corner of the regeneration grid. Seed traps were deployed in the spring and their contents were recovered in late autumn. White spruce seeds were identified and counted in the laboratory.

2.3. Data analysis

We calculated a substrate suitability index to determine the suitability of different substrates for the establishment of tree seedlings. To calculate the index, we divided the relative abundance of seedlings (by species and by height class) that were present on a given substrate by the percentage area that was covered by this same substrate (Côté and Bélanger, 1991; McLaren and Janke, 1996; LePage et al., 2000; Parent et al., 2003). We standardized the index by dividing its value for a particular substrate by the summed values for all substrates. Consequently, the standardized index ranged from 0 to 1, i.e., from low to high suitability of substrates for seedling establishment. To evaluate spatial dispersion of seedlings, we calculated their

stocking by dividing the number of subplots within a regeneration grid or along a transect with a least one seedling by the total number of subplots in the grid or transect. Seedling stocking was calculated by species and height class. For the purposes of data description, we considered regeneration that was >30 cm in height as likely to be established (Côté and Bélanger, 1991; Ruel et al., 1995; Westerberg, 1995; Lieffers et al., 1996b).

We analyzed density, composition, stocking, substrate availability, and the substrate suitability index with linear mixed-effects models (PROC MIXED; SAS Institute, 2003) with type of ecosystem and substrate as fixed effects, and regeneration grid and transects as random effects. We performed pairwise Tukey tests to verify differences between means. The Tukey method adjusts the P -values of the numerous tests to preserve an overall 5% error rate.

3. Results

3.1. Stand characteristics and seed availability

The structure of mature second-growth white spruce stands and overmature balsam fir stands both exhibited a unimodal DBH distribution of live stems, which were centered respectively on the 17 and 20 cm DBH classes (Fig. 3a and b). Consequently, both stand types had a similar density of live stems ($t_{1,8} = 0.11$, $P = 0.914$). DBH distributions of dead stems in both stand types were positively skewed given higher densities of small dead stems. Overmature balsam fir stands contained large dead stems up to the 50 cm DBH class, while the largest dead stems in the white spruce stands were in the 29 cm DBH class (Fig. 3c and d). Accordingly, large dead stems (DBH > 20 cm) were 5 times more abundant in overmature balsam fir stands ($200 \pm 46 \text{ stems ha}^{-1}$) than in mature second-growth white spruce stands ($42 \pm 38 \text{ stems ha}^{-1}$; $t_{1,8} = -2.65$, $P = 0.029$).

Overmature balsam fir stands were composed of balsam fir (80%), white spruce (18%) and paper birch (2%). Dead stems consisted of balsam fir (87%), white spruce (7%), paper birch (3%), and unidentified coniferous species (because of advanced decay; 3%). Mature second-growth white spruce stands were composed almost exclusively of white spruce (live, 98%; dead, 92%). White spruce seeds were twice as abundant in white spruce stands as in balsam fir stands (Fig. 3a and b: $t_{1,5} = 2.84$, $P = 0.036$).

3.2. Seedling density and stocking

Density and stocking of white spruce seedlings that were >30 cm in height (i.e., likely established) was very low in the understory of mature second-growth white spruce stands and in recent clear-cuts of mature second-growth white spruce stands, while it was significantly higher in overmature balsam fir stands, particularly in canopy gaps (Table 1). In balsam fir canopy gaps, densities and stocking of white spruce seedlings in the 31–60 cm height class were respectively 14- and 9-fold higher compared to the understory of mature second-growth white spruce stands (density, $t_{3,29} = -5.6$, $P < 0.001$; stocking, $t_{3,29} = -5.4$, $P < 0.001$). The differences in seedling density and stocking between these two stand types increased for the 61–130 cm height class to 28 times (density, $t_{3,29} = -5.0$, $P < 0.001$) and 14 times higher (stocking, $t_{3,29} = -5.3$, $P < 0.001$) in the former compared to the latter sites. Likewise, density and stocking of spruce seedlings in the 31–60 cm height class were respectively 5 and 3 times higher within balsam fir canopy gaps than in recent clear-cuts of mature second-growth white spruce stands (density, $t_{3,29} = 4.5$, $P < 0.001$; stocking, $t_{3,29} = 4.2$, $P < 0.001$); density and stocking of seedlings in the 61–130 cm height class were greatly elevated, in that they were respectively 34 and 27 times higher in the canopy gaps com-

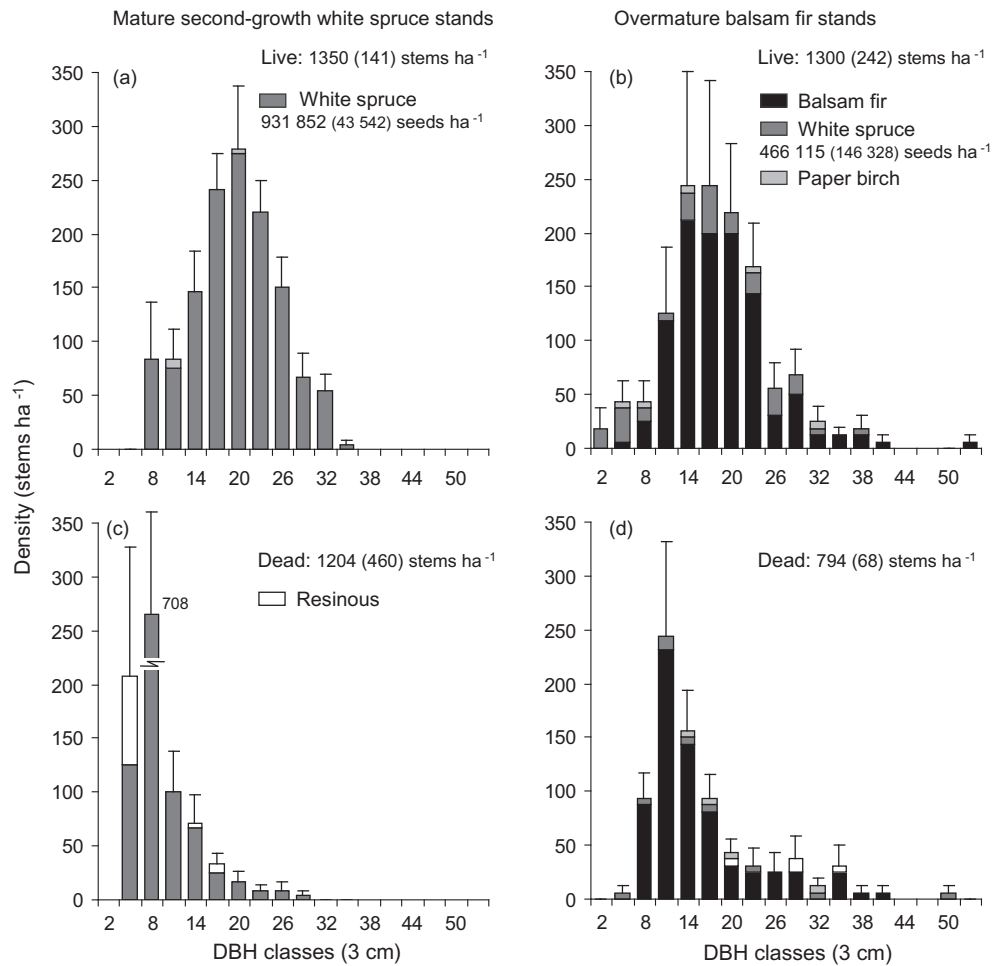


Fig. 3. Distribution of mean (± 1 SE) stem density by diameter at breast height (DBH) classes and mean (± 1 SE) seed density in 4 stands by cover type.

pared to the clear-cuts (density, $t_{3,29} = -5.0$, $P < 0.001$; stocking, $t_{3,29} = 5.3$, $P < 0.001$).

Density and stocking of white spruce seedlings in the 11–30 cm height class was higher in balsam fir canopy gaps (Table 1) than in both the understory of mature second-growth white spruce stands (7 times higher density, $t_{3,29} = -4.7$, $P < 0.001$; 3 times higher stocking, $t_{3,29} = -5.3$, $P < 0.001$) and in recent clear-cuts of mature second-growth white spruce stands (7 times higher density, $t_{3,29} = -5.0$, $P < 0.001$; 4 times higher stocking, $t_{3,29} = -5.6$, $P < 0.001$).

Density and stocking of white spruce seedlings in the 0–10 cm height class were high across all ecosystem types, except for recent clear-cuts of mature second-growth white spruce stands. In these latter stands, densities were 76 times lower ($t_{3,29} = -5.0$, $P < 0.001$) than in balsam fir canopy gaps, while stocking was 20 times lower ($t_{3,29} = -9.9$, $P < 0.001$).

Seedlings of balsam fir and paper birch ≤ 10 cm were abundant in overmature balsam fir stands but rare in mature second-growth white spruce stands and recent white spruce clear-cuts (Table 1). Seedlings of balsam fir and paper birch > 10 cm in height were rare in all four types of ecosystems.

3.3. Substrate suitability and availability for seedling establishment

The most suitable substrate for the establishment of white spruce seedlings is Rotten Log + Moss, closely followed by Rotten Log + Moss + Needles (Fig. 4). All other substrates, including min-

eral soil and every form of log without moss, are less suitable than Rotten Log + Moss ($t_{8,316} \geq -3.28$, $P < 0.001$). The least suitable substrate for establishment is Recalcitrant Layer Species. The suitability index of most substrates tended to decrease with increasing height of seedlings. This tendency is significant only for Rotten Log + Moss ($f_{8,316} = 4.38$, $P = 0.005$) and Moss + Needles ($f_{8,316} = 5.00$, $P = 0.002$). We have not shown results by ecosystem type because it did not influence the suitability of substrates ($f_{8,316} = 0.94$, $P = 0.422$). Substrates for which we have not shown an index in Fig. 4 (6 out of 15) were not abundant (Table 2) and had low suitability indices (< 0.03).

The most suitable substrate for white spruce seedling establishment (i.e., Rotten Log + Moss) was ~ 4 times less abundant in mature second-growth white spruce stands ($t_{3,29} = -5.41$, $P < 0.001$) and ~ 6 times less abundant in recent clear-cuts of mature second-growth white spruce stands ($t_{3,29} = -5.87$, $P < 0.001$) compared to balsam fir canopy gaps (Table 2). The ground surface of mature second-growth white spruce stands and recent clear-cuts of mature second-growth white spruce stands is mainly covered by substrates of low suitability. Needles, Moss and Moss + Needles substrates account for 89% of substrate area in mature second-growth white spruce stands, while the Recalcitrant Layer Species cover 85% of substrate area in recent clear-cuts of mature second-growth white spruce stands. In the recent clear-cuts, heights of live individuals within the Recalcitrant Layer Species group ranged between 73 cm and 102 cm, while 96% of white spruce seedlings are < 61 cm in height (Table 1).

Table 1Mean (± 1 SE) density (D , seedling ha^{-1}) and stocking (S , %) of seedlings in the four ecosystem types.

Ecosystem type	Species	Height classes of seedlings (cm)															
		Cotyledon		0–10		11–30		31–60		61–130			S ^a				
		D	S	D	S	D	S	D	S	D	S	D		S			
Understory of mature white spruce stand ^b	Balsam fir	5	(3)	<1	663	(332)	15	3	(3)	<1	0			0			15
	White spruce	408	(132)	11	6828	(1037)	60	1100	(312)	18	208	(79)	4	80	(56)	2	66
	White birch	0			190	(155)	3	10	(8)	<1	0			0			3
Recent clearcut of mature white spruce stand ^b	Balsam fir	0			0			8	(4)	<1	0			0			<1
	White spruce	0			88	(38)	3	851	(333)	15	558	(156)	11	67	(21)	1	22
	White birch	0			0			3	(3)	<1	0			0			<1
Understory of overmature balsam fir stand	Balsam fir	29,674	(11,305)	70	111,174	(35,761)	98	953	(364)	16	6	(6)	<1	6	(6)	<1	99
	White spruce	1316	(577)	22	8475	(1979)	68	2307	(662)	35	569	(287)	11	395	(278)	8	81
	White birch	1667	(960)	17	6020	(843)	42	106	(24)	3	6	(6)	<1	0			45
Gap in overmature balsam fir stand	Balsam fir	20	(20)	<1	53,954	(6492)	90	2997	(1351)	30	206	(206)	3	63	(63)	1	90
	White spruce	7	(7)	<1	6684	(1166)	59	6239	(1381)	59	2964	(702)	37	2267	(594)	27	83
	White birch	7	(7)	<1	4518	(1059)	36	172	(70)	4	19	(19)	<1	0			37

^a Mean stocking from all seedling height classes.^b Second-growth.

4. Discussion

4.1. Regeneration processes in mature second-growth white spruce stands

Density and stocking of established white spruce regeneration (i.e., >30 cm tall; Ruel et al., 1995; Westerberg, 1995; Lieffers et al., 1996b) was lower in mature second-growth white spruce stands compared to overmature balsam fir stands. In eastern North America, no regeneration standard exists for white spruce that could be used as a baseline for evaluating regeneration failure on Anticosti Island, given that white spruce rarely forms pure, even-aged stands under natural conditions in eastern North America (Stiell, 1976; Bell et al., 1990; Saucier et al., 2009), except in coastal, mountainous or riparian environments (Nienstadt and Zasada, 1990; Timoney and Peterson, 1996; Villeneuve et al., 2001). According to regeneration standards for western Canada, where white spruce stands have a broader natural distribution on mesic sites (Nienstadt and Zasada, 1990), stem densities should be >4000 individuals ha^{-1} for seedlings that are >50 cm in height pre-harvest to achieve full stocking (i.e., 80%) post-harvest (Lieffers and Beck, 1994; Greene et al., 2002). Mature second-growth white spruce stands in Anticosti Island only contained 288 white spruce seedlings ha^{-1} that were >30 cm tall, with a stocking of 5%. According to Kneeshaw et al. (2000), stocking is more important than density in ensuring stand replacement after disturbance. Thus, it was highly unlikely that mature second-growth white spruce stands will recover from eventual clear-cutting and return to a forest state.

Regeneration failure in mature second-growth white spruce stands could not be attributed to a lack of seeds since seed trees and seeds were more abundant on these sites than in overmature balsam fir stands. Furthermore, seed density in mature second-growth white spruce stands was about twofold higher than what is considered necessary to reach full stocking from sowing

(Greene et al., 2002). Rather, regeneration failure in mature second-growth white spruce stands could be explained by a lack of suitable substrates for establishment of white spruce seedlings. The forest floor was dominated by substrates of low suitability, whereas the most suitable substrate was rare. Indeed, Rotten Log + Moss was 4 times less abundant in mature spruce stands than in balsam fir canopy gaps. Moreover, large dead trees that were susceptible to creating suitable rotten logs were 5 times less abundant. Moss on rotten logs is largely recognized as one of the most suitable substrates for white spruce seedling establishment (Côté and Bélanger, 1991; Lieffers et al., 1996b; Simard et al., 1998; LePage et al., 2000; Peters et al., 2006). The great suitability of rotten logs can be explained by their elevated position, which allows litter to be shed easily, together with their porous structure that facilitates root development, and provides higher temperatures, greater moisture retention, improved nutrition, and a more stable water supply than the underlying soil layers (Rowe, 1955; Delong et al., 1997). Notably, white spruce requires a microenvironment that is supplied with continuous moisture to survive the early seedling phase of establishment (Hogg and Schwarz, 1997). We noted that mosses represented a suitable substrate for seedling establishment when in combination with rotten logs. Mosses probably maintained high moisture content in the logs by isolating the latter from direct sunlight. Mosses may also have acted as traps for seeds, maintaining these on logs.

Mature second-growth white spruce stands had not yet reached the stand break-up stage occurring in overmature stands, during which canopy gap mortality naturally starts producing large dead trees (McCarthy, 2001). Hence, the younger developmental stage of mature second-growth white stands can explain the lower number of logs than was observed in overmature balsam fir stands. In overmature white spruce stands, logs may not yet be suitable habitats for regeneration establishment (i.e., few rotten logs). Timoney and Peterson (1996) showed that clear-cutting of old-growth riparian white spruce stands also triggered regeneration failure. In their

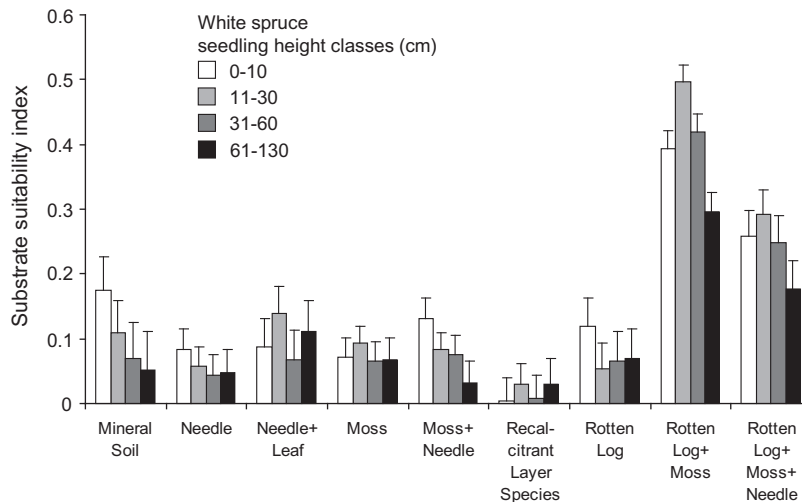


Fig. 4. Estimated ($\bar{X} \pm 1$ SE) index of the suitability of nine substrates for establishment of white spruce seedlings. The index ranges from 0 (low suitability) to 1 (high suitability).

study, old-growth white spruce stands (~210-years-old) contained insufficient regeneration for them to recover from clear-cutting. Stocking (28%) and density (960 stems ha^{-1}) of likely established white spruce seedlings were lower than both values that were estimated for overmature balsam fir stands on Anticosti Island and regeneration standards for western Canada (Lieffers and Beck, 1994; Greene et al., 2002). The slower decomposition rate of white spruce compared to balsam fir (Whitney, 1989; McCarthy, 2001) may also have explained the lower number of suitable rotten logs in mature second-growth white spruce stands compared to the overmature balsam fir stands. Because of their slower decomposition rates, white spruce logs likely become suitable rotten logs less rapidly than do balsam fir logs. Moreover, white spruce windfalls can remain perched on their branches for an unknown length of time, which will inevitably delay the stage at which the bole touches the ground and reaches a decay state and moisture levels that are suitable for establishment of white spruce seedlings (unpublished field observations). Overmature white spruce stands may eventually contain similar numbers of large logs as overmature balsam fir stands, but they may not become suitable rotten logs rapidly enough for the establishment of a dense seedling bank that would enable the stand to recover from natural canopy gap mortality, i.e., mortality rates could be higher than regeneration rates. Thus, overmature white spruce stands could gradually lose canopy cover density over time and never reach regeneration levels that would enable the stand to recover from clear-cutting.

4.2. Regeneration processes in recent clear-cuts of mature second-growth white spruce stands

In recent clear-cuts of mature second-growth white spruce stands, density and stocking of white spruce regeneration was even lower than in mature second-growth white spruce stands. Moreover, the least suitable substrate for establishment of white spruce seedlings covered 85% of the area, i.e., Recalcitrant Layer Species. Growth of these understory species overtopped 96% of all white spruce seedlings. White spruce seedlings are vulnerable to overtopping by competing vegetation which subsequently impedes their establishment (Cole et al., 2013). Recalcitrant understory layer species, most notably *C. canadensis*, can completely colonize a site within 3 years after disturbance, and go on to produce a thick litter layer that shades and insulates the soil, thereby inhibiting the establishment and growth of white spruce seedlings

(Hogg and Lieffers, 1991; Lieffers et al., 1993; Royo and Carson, 2006). Because of a lack of white spruce regeneration and the encroachment by recalcitrant understory layer species which exacerbate the lack of suitable substrates of establishment, it is highly unlikely that recent clear-cuts of mature second-growth white spruce stands will recover from clear-cutting and return to a forest state. Hence, the cumulative effects of chronic deer browsing and clear-cutting of mature second-growth white spruce stands have altered regeneration processes and triggered an alternative successional pathway toward the formation of parkland, i.e., partial deforestation. This novel ecosystem has undoubtedly compromised biodiversity (e.g., changes in ecosystem composition and function) and ecosystem services (e.g., lower carbon sequestration and wood fiber production) that had been provided by the preindustrial forest (Barrette et al., 2010; Bridgewater et al., 2011). Moreover, parklands could persist despite reductions in deer populations. Effectively, the recalcitrant understory layer species that contribute to the formation of these parklands are the persistent legacies following reductions in deer densities (Nuttall et al., 2014).

4.3. Conclusion

Regeneration processes of mature white spruce stands have not appeared to enable white spruce stand recovery from clear-cutting, most probably because of their inability to produce sufficient numbers of suitable rotten logs for the establishment of a dense understory that is comprised of their own regeneration. This new hypothesis could be tested in a future experiment isolating the effects of recalcitrant understory layer species and in which white spruce and balsam fir logs are added to mature second growth white spruce stands before cutting. Other studies have also recognized that sufficient quantities of decayed wood are necessary for forest regeneration (Hofgaard, 1993; Kuuluvainen and Juntunen, 1998; Uotila et al., 2002). From this perspective, the capacity of balsam fir logs to rot rapidly can help ensure forest resilience. We believe that this characteristic allows balsam fir to produce sufficient numbers of suitable rotten logs that can facilitate regeneration processes. Thus, the occurrence of white spruce stands on Anticosti Island are probably not attributable to regeneration processes originating in white spruce stands but to a regeneration legacy of balsam fir stands. White spruce stands do occur naturally in western North America, but they are considered a highly transitional serial stage between trembling aspen (*Populus tremuloides*

Table 2Mean percentage (± 1 SE) of the area covered by seedling substrates and mean heights of logs and live individuals of Recalcitrant Layer Species by ecosystem type.

Substrate of establishment	Understory of mature white spruce stand ^a		Recent clearcut of mature white spruce stand ^a		Understory of overmature balsam fir stand		Canopy gap in overmature balsam fir stand	
	Substrate area (%)	Height (cm)	Substrate area (%)	Height (cm)	Substrate area (%)	Height (cm)	Substrate area (%)	Height (cm)
Mineral soil	0		<1	(<1)	<1	(<1)	1	(<1)
Needle	18	(3)	<1	(<1)	28	(3)	13	(3)
Leaf	0		0		<1	(<1)	1	(1)
Needle + Leaf	<1	(<1)	0		8	(4)	7	(2)
Moss	9	(3)	1	(<1)	17	(7)	20	(6)
Dry Moss	0		3	(1)	0		0	
Moss + Needle	62	(5)	0		30	(2)	31	(2)
Rock	0		<1	(<1)	0		0	
Fresh Log	1	(<1)	0		<1	(<1)	<1	(<1)
Dry Log	1	(1)	8 ^b	(2)	6	(1)	4	(2)
Rotten Log	0		0		1	(<1)	1	(1)
Rotten Log + Moss	3	(1)	2	(1)	6	(1)	11	(2)
Rotten Log + Moss + Needle	5	(2)	0		3	(2)	4	(2)
Rotten Log + Dry Moss	0		1	(1)	0		0	
Recalcitrant Layer Species	1	(<1)	85	(2)	2	(2)	13	(3)
<i>Calamagrostis canadensis</i>	1	(<1)	62	(4)	2	(1)	13	(3)
<i>Pteridium aquilinum</i>	<1	(<1)	1	(<1)	<1	(<1)	1	(1)
<i>Cirsium</i> spp.	0		2	(<1)	0		0	
Mixed together	1	(<1)	12	(2)	0		<1	(<1)
Mixed with other substrates	1	(<1)	8	(2)	0		<1	(<1)

^a Second-growth.^b Represents slash left by the harvester.

Michaux) stands that originate from fire and old-growth balsam fir stands (Liefvers et al., 1996a; Liefvers et al., 2003). These white spruce stands are usually regenerated through site preparation and planting after harvest, and at considerable cost (Liefvers et al., 2008).

4.4. Implications for management of white spruce stands

Because of deer browsing, balsam fir is no longer present to facilitate regeneration processes in the white spruce stands of Anticosti Island. In this context, we have shown that maintenance of white spruce stands following clear-cutting may be very difficult without planting. Shelterwood cuts that expose mineral soil could promote white spruce regeneration in white spruce stands and reduce encroachment by a recalcitrant ground cover layer (Raymond et al., 2000; Liefvers et al., 2003). Yet, the presence of exposed mineral soil is temporally limited because the growth of substrates of low suitability (e.g., mosses and coniferous litter) will eventually cover the exposed mineral layer. Hence, the success rate of this silvicultural alternative remains uncertain because it is dependent upon timing mineral soil exposure with a good seed year, which is a highly variable occurrence (2–12 years; Nienstadt and Zasada, 1990). Moreover, Timoney and Robinson (1996) showed that the preferred rooting medium of white spruce shifts early in succession from mineral soil to decaying wood under pre-existing canopies. Thus, exposed mineral soil was not a particularly favorable substrate for white spruce seedling establishment under the canopy of both stand types, while moss growing on rotten logs was by far the most suitable.

Shelterwood cuts that create nurse logs (i.e., rotten logs suitable for regeneration establishment) could be investigated. To ensure sufficient white spruce regeneration, suitable rotten logs should cover $\geq 11\%$ of the substrate area (present study) in canopy openings that are $>200 \text{ m}^2$ in size. Targeting smaller openings for regeneration is not recommended because these can trigger the development of an alternative successional pathway toward parklands in overmature balsam fir stands (Barrette et al., unpublished manuscript). We did not measure the volume of logs that was necessary to cover $\geq 11\%$ of the substrate area. Dead wood

(diameter $\geq 10 \text{ cm}$) in unmanaged old-growth balsam fir stands can range between $40 \text{ m}^3 \text{ ha}^{-1}$ and $94 \text{ m}^3 \text{ ha}^{-1}$ (Côté et al., 2009). This would represent a considerable volume left upon the ground considering that merchantable volume (DBH $\geq 9.1 \text{ cm}$) of old-growth balsam fir stands can be around $200 \text{ m}^3 \text{ ha}^{-1}$ (Tremblay et al., 1997; McCarthy and Weetman, 2007b) and that a continuous cover shelterwood system should harvest $\geq 66 \text{ m}^3 \text{ ha}^{-1}$ every 20–30 years to be economically sustainable (i.e., 33% of merchantable volume; Raymond et al., 2013). Moreover, if the species of the logs does not have the capacity to rot rapidly (e.g., white spruce), they may not produce white spruce regeneration rapidly enough for the stand to recover from repeated shelterwood cutting. In conclusion, shelterwood cutting that create nurse logs raises diverse questions (e.g., economic, ecological). Hence, further research is needed to test our new hypothesis that regeneration failure is due mainly to the absence of rotten logs and to determine if pure white spruce stands can be maintained without plantations.

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